Mayank Goud Polagoni

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Education

Northeastern University, Boston, MA May 2026

Master of Science in Robotics, ***Concentration :*** *Electrical and Computer Engineering*

Relevant Coursework: Robotic Sensing and Navigation, Mobile Robotics, Control Systems Engineering, Robot Mechanics and Control, Reinforcement Learning, Pattern Recognition and Computer Vision

CVR College Of Engineering, Hyderabad, India 2019 – 2023

Bachelor of Technology in Electrical and Electronics Engineering

Technical Skills

Programming & Tools: Python, C++, MATLAB, Simulink, ROS / ROS 2, Linux, Git, Docker, Gazebo, LaTeX

Computer Vision & Perception: OpenCV, YOLOv8, MediaPipe, SIFT, ORB, Harris Corners, RANSAC, Camera Calibration & AR, Multi-Object Tracking, Pattern Recognition

Deep Learning & Reinforcement Learning: PyTorch, NumPy, SciPy, Scikit-learn, CNN, Transfer Learning, PPO, SAC, TD3, DDPG, REINFORCE, Policy Gradient Methods, Actor-Critic, Q-Learning, MDPs, OpenAI Gymnasium

Robotics, SLAM, Estimation & Control: RTAB-Map, Visual SLAM, MPPI, Pure Pursuit, Gaussian Process, Kalman Filter, Sensor Fusion, Allan Variance, PID & State-Space Control, Lyapunov Stability, LiDAR, IMU, Jetson Orin NX

Projects

Deep Reinforcement Learning for Autonomous Racing in F1TENTH **Jan 2026 – Apr 2026**

Northeastern University

- Implemented PPO, SAC, and TD3 deep RL algorithms in PyTorch for autonomous racing in the F1TENTH Gym simulator using continuous LiDAR-based observations and Ackermann steering dynamics
- Designed reward functions and Gymnasium-compatible environment wrappers, conducting systematic comparison against classical baselines (Pure Pursuit, reactive LiDAR controllers) across five tracks
- Achieved 2× the reward of the best classical baseline with SAC (avg reward 0.289, avg speed 2.89 m/s) and a peak speed of 3.0 m/s with TD3, showing clear off-policy superiority over PPO
- Conducted multi-seed stability analysis and reward-shaping ablations, evaluating generalization to an unseen Berlin track and reporting findings in an AAAI-format report

Real-Time Multi-Object Tracking with Custom Data Association **Jan 2026 – Apr 2026**

Northeastern University

- Built a real-time multi-object tracking pipeline from scratch, combining YOLOv8n detection with a custom Kalman Filter for motion prediction and the Hungarian algorithm for data association
- Implemented two cost matrices (IoU and Euclidean distance) for assignment, enabling empirical comparison of matching strategies under occlusion and fast-motion scenarios
- Achieved real-time throughput of 5.9–8.9 FPS on a consumer laptop while maintaining stable identity assignment across frames, with graceful handling of partial occlusion

RTAB-Map Onboard COBRA **Sep 2024 – Dec 2024**

Northeastern University

- Implemented RTAB-Map SLAM on COBRA (Crater Observing Bio-inspired Rolling Articulator) robot using RGB-D cameras and Jetson Orin NX for real-time mapping
- Optimized SLAM performance by benchmarking odometry against OptiTrack ground truth, enhancing system accuracy for challenging sidewinding gaits
- Analyzed and addressed sparse point cloud issues in low-object environments by evaluating RTAB-Map's limitations and proposing LiDAR integration for future enhancements
- Collaborated within an interdisciplinary team to collect and process experimental data in varying environments, ensuring robust evaluation of mapping algorithms

Work Experience

Defence Research and Development Laboratory (DRDL) Oct 2022 – Jan 2023

Electrical Engineering Intern Hyderabad, India

- Developed a MATLAB state feedback controller , utilizing analytical thinking to conduct parameter analysis and enhance control accuracy
- Applied pragmatic problem-solving to implement advanced control strategies, improving actuator efficiency and dynamic response
- Implemented advanced control strategies like pole placement and adaptive tuning to improve actuator efficiency, reduce energy consumption, and enhance dynamic response
- Refined closed-loop control algorithms to minimize oscillations , achieving a 35% stability improvement